Searching for Type Ia Supernovae in ZTF Deep Fields

Development of an Open-Source Detection Framework

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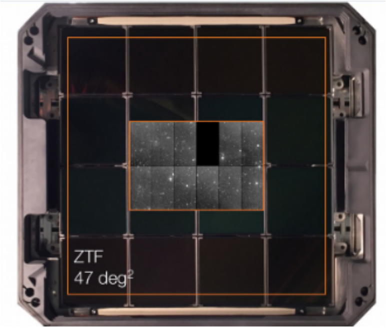
Outline

Scientific Context

The Zwicky Transient Facility (ZTF)

- Located at Palomar Observatory (California)
- 48-inch Samuel Oschin Telescope
- Mosaic of 16 CCDs, each with 4 readout channels (quadrants)
- Covers 47 deg^2 per exposure
- Observes in g, r, and i filters
- Limiting magnitude: ≈ 20.5 (30s exposures)



ZTF camera installed on the telescope (Abbot et al. 2017).

Key point: ZTF scans wide fields nightly, but faint transients can be missed by real-time pipelines.

Type Ia Supernovae in Cosmology

- Standard candles → extragalactic distances
- Discovery of cosmic acceleration (Nobel Prize 2011)
- Central to the Hubble constant tension

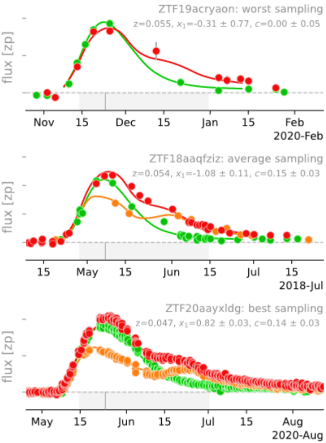
Key point: Type Ia supernovae are remarkably uniform, and visible across cosmological distances.

SN 2011fe in the Pinwheel Galaxy (M101), observed by the Palomar Transient Facility.

Credit: B.J. Fulton / LCOGT

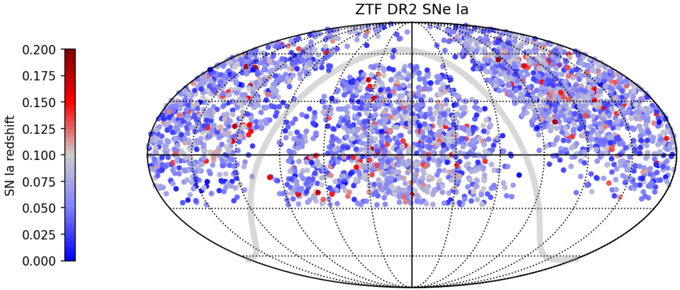
Light Curve of Type Ia Supernovae

- Thermonuclear explosion of a white dwarf near the Chandrasekhar limit
- Produces a characteristic light curve: sharp rise, gradual decline
- Brightness–decline relation → usable as a standard candle



Lightcurves from ZTF SN Ia DR2: Overview.

Overview of the ZTF-COSMO-DR2 Catalog



- Public catalog of 3,628 spectroscopically confirmed SNe Ia
- Observation window: March 2018 to January 2021
- Includes position, redshift, discovery date, classification, photometry
- Built from real-time alerts + crossmatching with external surveys
- Used as a reference for evaluating missed detections

Key point: Our stacking approach could increase detection density, especially in low-exposure fields.

Scientific Goal

Objective

Demonstrate that stacking archival ZTF images can improve the detection of faint Type Ia supernovae.

Hypothesis

Stacking images increases the limiting magnitude, allowing more SNe Ia to emerge above the detection threshold.

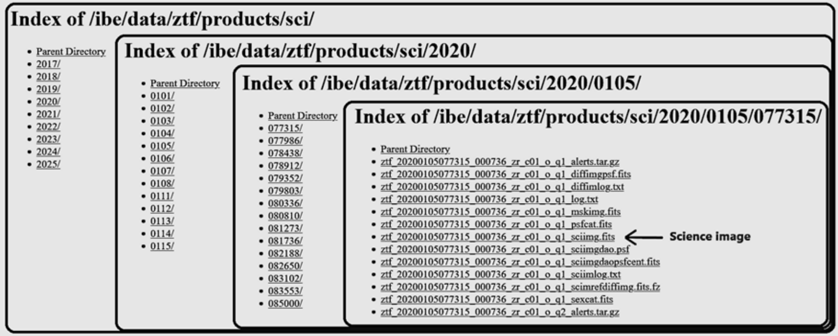
Strategy

Download public science images, align and normalize them, then perform temporal stacking followed by subtraction from a median reference.

Key point: Reprocessing existing public data can reveal transients missed by the original pipeline, without any new observations.

Data Structure and first results

How Public ZTF Data is Structured



Example of IRSA file hierarchy and fractime structure.

- Data hosted at IRSA in HTML format
- Files are organized by date, not sky region
- $\approx 544,000$ index pages to scan (2018-2021)

Building a Queryable CSV Index

Field

	A	B	C	D	E	F	G
1	Column1	Column2	Column3	Column4	Column5	Column6	Column7
2	199	200	201	202	203	204	205
3	20200912391979	20200912389352	20200916388264	20190815488356	20201006335394	20200902438345	20200903458854
4	20200914327477	20200916355752	20200917349097	20201003313194	20201006396505	20201006355093	20201004396343
5	20200920301644	20200916376528	20200917393854	20201003377106	20201012377014	20201006406458	20201004401192
6	20200920360926	20200917333877	20200919355000	20201006313021	20201014313252	20201008372847	20201006411620
7	20201006250532	20200917376447	20200919397303	20201006314109	20201014354826	20201008418206	20201006418125
8	20201006292199	20200918334213	20200921334155	20201006377164	20201016314433	20201012397801	20201008375648
9	20201008251713	20200918356111	20200921375463	20201006390856	20201016356204	20201014352373	20201008436956
10	20201012250822	20200920313044	20201006301759	20201008355984	20201018304850	20201016333854	20201009394606
11	20201012292998	20200920385764	20201008293241	20201012313333	20201018356377	20201018355775	20201009418611
12	20201014230081	20200921313206	20201008357975	20201014351435	20201020356933	20201018376574	20201012396736
13	20201014293403	20200921355706	20201012271620	20201016313356	20201022333900	20201020373160	20201012418275
14	20201016250486	20201006272292	20201012314780	20201016355602	20201024333935	20201020376632	20201014375590
15	20201016272743	20201008271516	20201014334815	20201018292581	20201027292558	20201023335370	20201018396400
16	20201018272188	20201013251424	20201016292268	20201022314282	20201027335012	20201027313310	20201020374352
17	20201020208796	20201015250648	20201019306968	20201025313275	20201029271435	20201028292338	20201022341609
18	20201022208993	20201017271377	20201021315255	20201028273646	20201029335081	20201029313345	20201022397315
19	20201024229919	20201019293079	20201023271424	20201029252569	20201030318229	20201029351331	20201024376713
20	20201027251354	20201022229792	20201025292326	20201030266030	20201031333044	20201030317292	20201028377569
21	20201028208866	20201024242998	20201028272581	20201031314387	20201101311562	20201031333981	20201029314387

Data : year_month_day_fractime : yyyymmddffffff

- Local parsing of 544,000 IRSA index pages
- One unified `.csv` file: field, date
- The CSV index makes ZTF data fully searchable and scriptable.

Key point: Before, it's take 90 min to get 1 image, now I get 3 years in this time !

Using the CSV with Python

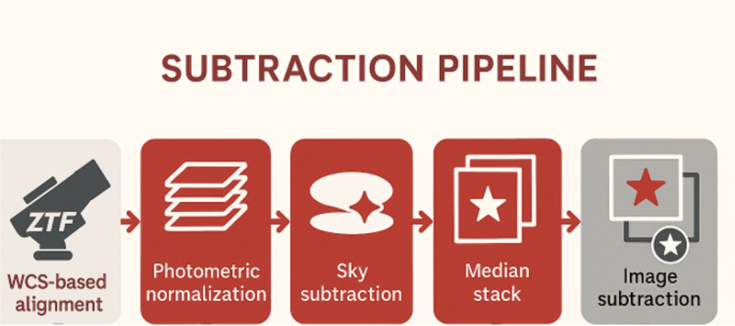
```
download_images_from_csv(
    csv_path="fichiers_par_field.csv",
    target_field="0625",
    filter_band="zr",
    ccd="06",
    quadrant="1",
    start_date="20200406",
    end_date="20210803")
```

Code snippet extracted from the CSV-Downloader module.

- Function `download_images_from_csv()`:
  - Selects only the observations needed
  - Filters by field, filter, CCD, quadrant, and date
  - Downloads only relevant science images
- Powered by the `fichiers_par_field.csv` file

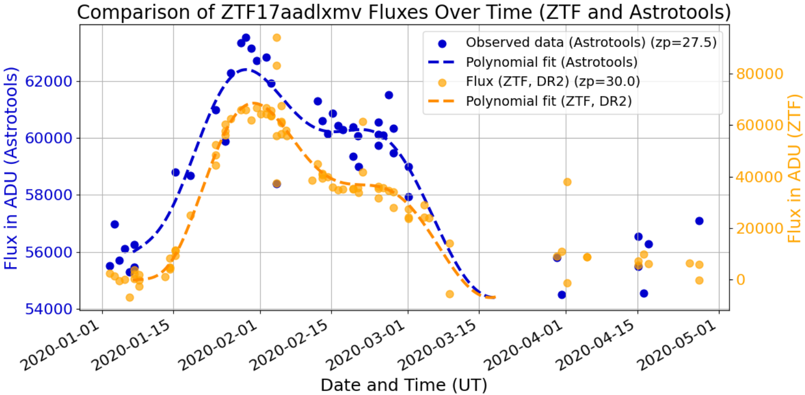
Processing Pipeline

AstroTools's processing pipeline



Pipeline based on Quentin Arvois's implementation.

Pipeline Validation



Light curve extracted with AstroTools (ZTF-r filter). Catalog comparison confirms accuracy.

- Confirmed Type Ia supernova detected in 2020
- Pipeline reconstructed its light curve
- Primary peak and secondary bump are clearly recovered

Practical Limitations for stacking

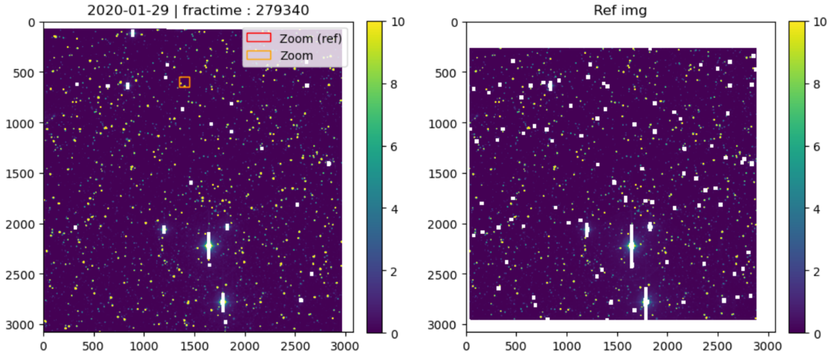
- Not all downloadable images are usable for stacking:
  - Poor seeing conditions
  - High sky background (moonlight, clouds...)
  - Instrumental artifacts or saturation

- Filtering criteria:
  - SEEING < 3.0 arcsec
  - Pixel median < 400 ADU

Key point: Real-world conditions reduce theoretical performance, but stacking still yields a clear advantage.

Deep Field Stacking and perspectives

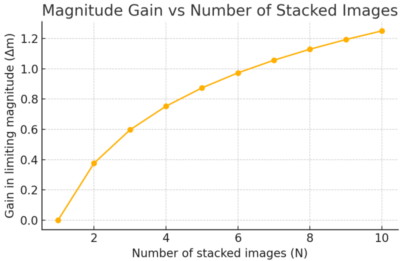
Subtraction Pipeline



Example of subtraction: the left stack contains 3 images, the right stack includes ≈ 350 images.

All images were selected and retrieved using a custom CSV-based indexing method.

Stacking and Detection Workflow



- Signal-to-noise improves as $\approx \sqrt{N}$
- Magnitude gain: $\Delta m = 1.25 \log_{10}(N)$
- For $N = 3$, gain is ≈ 0.6 mag
- Enables detection of fainter SNe Ia in archival data

Key point: Even small stacks significantly increase depth, revealing transients missed in real time.

Expected Gain in SNe Ia Detection

- **Observation period:** 828 nights (2018–2021)
- **ZTF fields:** 1800 fields $\times$ 47 deg^2 $=$ 84,600 deg^2
- **Usable images:** $\approx$50% after seeing and background filtering
- **Stacking gain (N=3):** $+$0.6 mag $\Rightarrow$ $\times 2.28$ more sources
- **Raw detection rate:** 5.1×10^{-5} SNe Ia / deg^2 / night

$$N_{\text{stack}} = 5.1 \times 10^{-5} \times 2.28 \times 84,600 \times 828 \times 0.5 \approx \textbf{4068 SNe Ia}$$

$$\Rightarrow \textbf{Estimated gain: } \boxed{4068 - 3628 = \textbf{440 additional SNe Ia}}$$

Key Point: Even with realistic filtering, stacking could reveal ≈ 440 new SNe Ia in public ZTF data.

Conclusion and Next Steps

Conclusion

- Archival ZTF data contains faint transients overlooked in real-time
- Temporal stacking boosts SNe Ia visibility by ≈ 0.6 mag
- The pipeline is validated on a known SN and generalizes to full fields
- Public and open-source: reproducibility, scalability, collaboration

Key point: This work demonstrates how to extract more science from what we already observe.

Next Steps and Perspectives

- Automate transient detection from residuals
- Implement confidence scoring and photometric classification
- Scale up to multiple fields in parallel
- Release a Python open-source library publicly and support contributions

Vision: Democratize SNe Ia discovery through open infrastructure.

Thank You

Thank you for your attention!

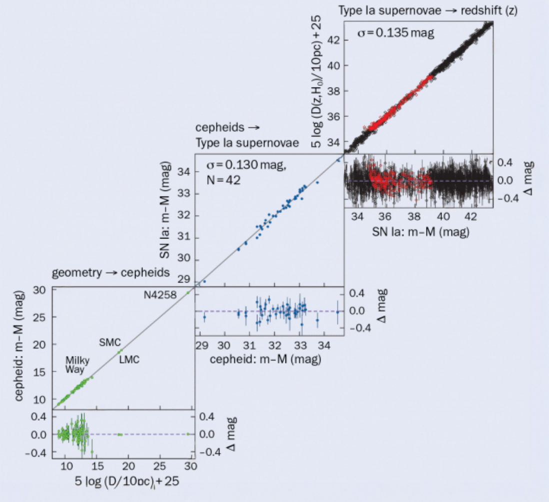
Questions?

All data used is publicly available.

Codebase is modular, reproducible, and (well be soon) open source.

Backup Slides

Hubble Tension and Dark Energy



Source : AURA (2023)

Hubble Tension & Cosmology with SNe Ia

- **Hubble tension**:
  - Local (SNe Ia + Cepheids): $H_0 \approx 73$
  - Early universe (CMB + ΛCDM): $H_0 \approx 67.4$
- *Discrepancy* $\approx 5\sigma \rightarrow$ *new physics?*